

Question 1: Stochastic Gradient Descent (SGD)

Problem Statement

Consider the following dataset for a simple linear regression problem:

$$x = [1, 2, 3, 4, 5]$$

$$y = [2, 4, 5, 4, 5]$$

Assume the linear regression model:

$$\hat{y} = wx + b$$

and the loss function for a single training sample:

$$L_i = \frac{1}{2}(y_i - \hat{y}_i)^2$$

where

$$\hat{y}_i = wx_i + b$$

The gradients for a single sample are:

$$\frac{\partial L_i}{\partial w} = -(y_i - \hat{y}_i)x_i$$

$$\frac{\partial L_i}{\partial b} = -(y_i - \hat{y}_i)$$

Implement **Stochastic Gradient Descent (SGD)** to estimate the parameters w and b of the linear regression model.

- Initial weight: $w_0 = 0$
- Initial bias: $b_0 = 0$
- Learning rate: $\eta = 0.01$
- Number of epochs: 1000

Tasks

1. Implement SGD in Python.
2. Calculate the prediction and loss for each sample.
3. Calculate the gradients of w and b .
4. Update w and b after every sample.

5. Calculate and store the Mean Squared Error (MSE) after every epoch.
Calculate the Mean Squared Error (MSE) after each epoch using:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

6. Plot the MSE versus epochs to observe convergence.
7. Display the final values of w and b .
8. Display the final MSE.